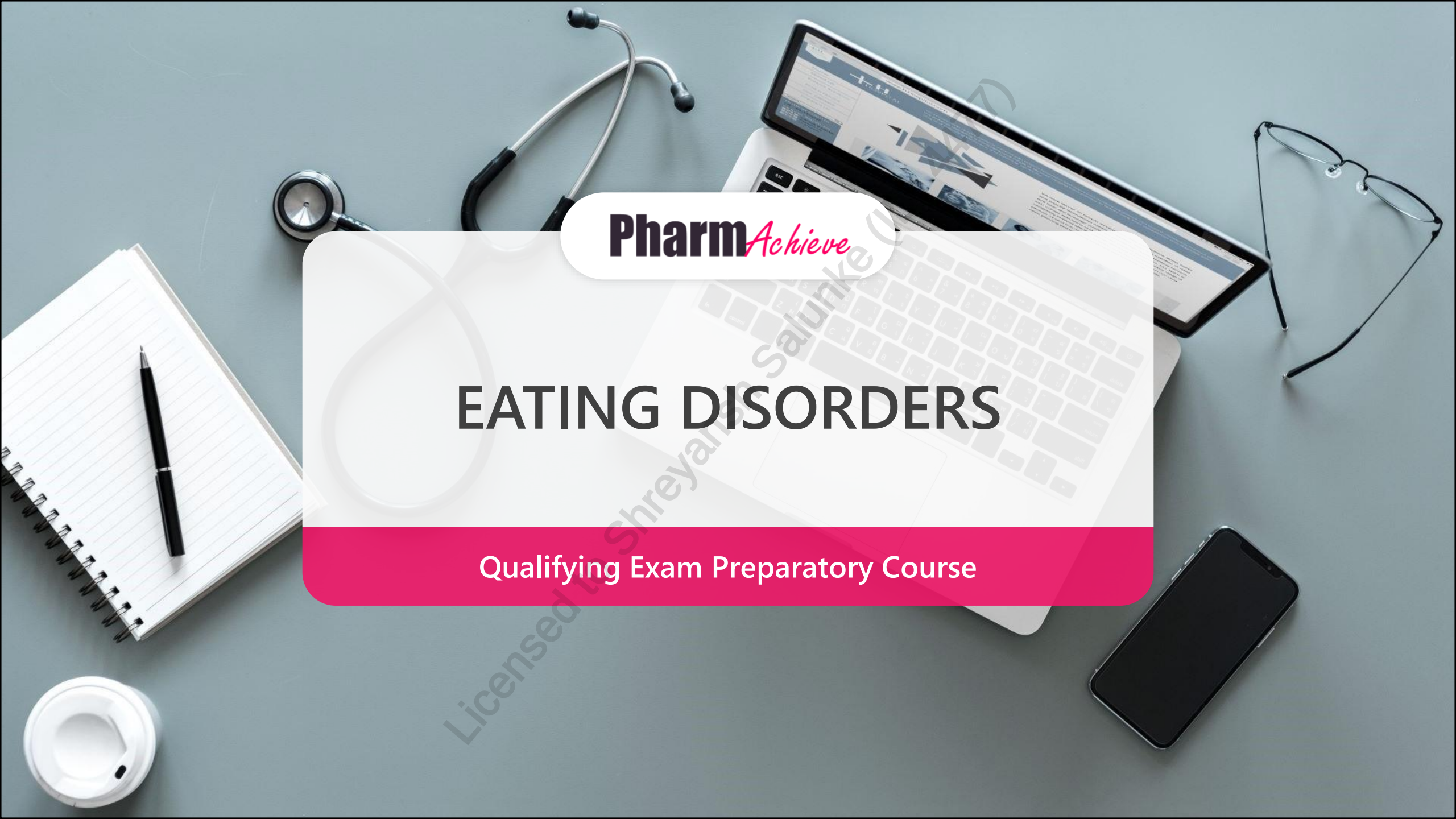


A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

EATING DISORDERS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

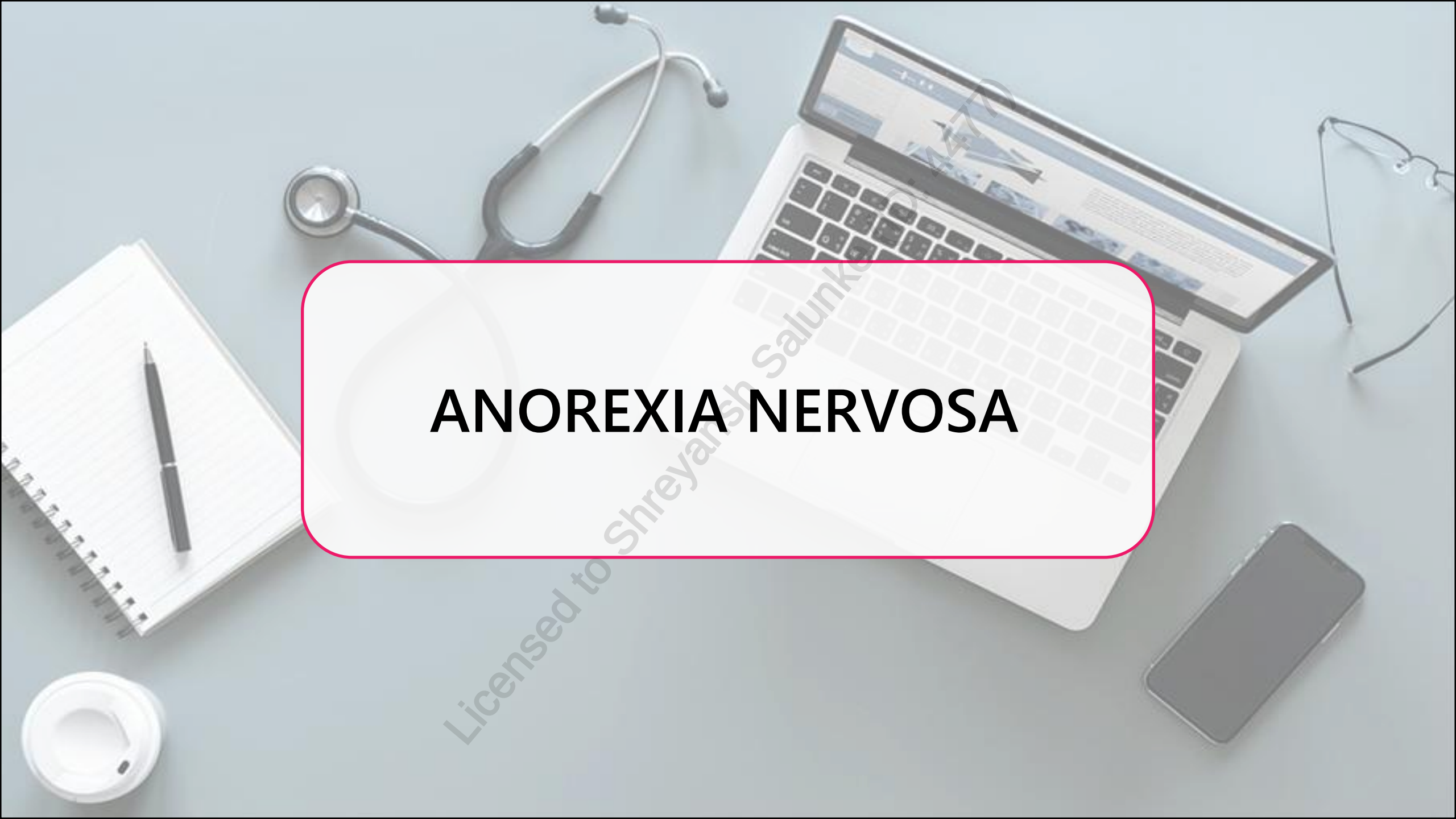
Professionalism

LEGEND

AEs	Adverse Effects	ECG	Electrocardiogram
BMI	Body Mass Index	EPS	Extrapyramidal Symptoms
BPM	Beats Per Minute	GABA	Gamma-amino butyric acid
[C]	Concentration	GI	Gastrointestinal
C/I	Contraindications	5-HT	Serotonin
CBT	Cognitive Behavioral Therapy	MAOI	Monoamine Oxidase Inhibitor
CBZ	Carbamazepine	MOA	Mechanism of Action
CNS	Central Nervous System	NGT	Nasogastric Tube
CYP	Cytochrome P450	NSAID	Non-Steroidal Anti-Inflammatory Drugs
D/I	Drug Interactions	OCD	Obsessive Compulsive Disorder
DRESS	Drug Reaction with Eosinophilia and Systemic Symptoms	SERT	Serotonin Transporter
DSM	Diagnostic and Statistical Manual of Mental Disorders	SSRI	Selective Serotonin Reuptake Inhibitor
		SNRI	Serotonin-Norepinephrine Reuptake Inhibitor
		TPN	Total Parenteral Nutrition

GOALS OF THERAPY

- ✓ Treat nutritional deficiencies and consequences of malnutrition (e.g. osteoporosis, hypoglycemia, dehydration)
- ✓ Minimize view of distortion on body and weight
- ✓ Reduce bingeing and purging behaviors
- ✓ Treat comorbid medical and psychiatric conditions
- ✓ Improve psychological function
- ✓ Establish healthy body weight and encourage healthy eating habits
- ✓ Prevent relapse

A top-down view of a light blue desk with various items: a silver stethoscope, a laptop displaying a medical website, a spiral notebook with a pen, a pair of glasses, a smartphone, and a white coffee cup. A large white rounded rectangle with a red border is centered over the laptop.

ANOREXIA NERVOSA

Licensed to Shreyansh Salunkhe (24477)

RISK FACTORS

- Family history of eating disorders
- History of physical or sexual abuse
- Concomitant mental health disorders (depression, anxiety, OCD)
- Low self-esteem
- Bullying
- Family dysfunction
- Gender (female > male)

*Diagnosis made by **DETAILED HISTORY, PHYSICAL EXAMINATION, LAB TESTS**
DSM-5 criteria used for diagnosis*

SUBTYPES

RESTRICTING TYPE

Over the last 3 months,
NO recurrent episodes of binge eating or
purging behaviors

Weight loss is primarily 'achieved' by:

- Restricting food intake (type, quantity)
- Excessive exercise

BINGE EATING/PURGING TYPE

Over the last 3 months, recurrent episodes
of binge eating or purging behaviors

Purging is defined as any of the following:

- Self-induced vomiting (mechanical or via misuse of emetic drugs)
- Misuse of laxatives
- Misuse of diuretics
- Misuse of enemas

CLINICAL PRESENTATION

- Deliberate restriction of food intake
- Intense fear of weight gain
- Persistent behaviors that aim to prevent weight gain (i.e. excessive exercise, etc.)
- Distorted perception of one's own body
- Denial of low body weight
- Dysphoria and low self-esteem
- Weakness, lethargy, syncope
- Low libido

SEVERITY

The DSM-5 classifies severity using the BMI

- Mild: $\geq 17 \text{ kg/m}^2$
- Moderate: $16\text{-}16.9 \text{ kg/m}^2$
- Severe: $15\text{-}15.9 \text{ kg/m}^2$
- Extreme: $< 15 \text{ kg/m}^2$

Other factors must also be considered, including but not limited to:

- Presence of suicidal ideation
- Functional impairment
 - Activities delayed or missed due to restrictive or bingeing/purging behaviours

COMPLICATIONS

ALL SUBTYPES

- Hypothermia
- Anemia
- Amenorrhea (absence of menstruation)*
- Bradycardia
- Hypotension
- Dizziness
- Osteoporosis
- Edema
- Dry skin
- Constipation
- Brittle hair, fine hair growth, or hair loss
- Refeeding Syndrome
- In very severe cases, anorexia can be fatal

* Note: patients CAN become pregnant even if not menstruating

SELF-INDUCED VOMITING

- Tooth enamel deterioration
- Dehydration
- Hypokalemia, hypochloremia, metabolic alkalosis
- Parotid hypertrophy
- Russell's sign (scars over knuckles)
- Mallory-Weiss tear

REFEEDING SYNDROME

- Serious, potentially fatal complication characterized by significant electrolyte disturbance
 - Hypophosphatemia, hypokalemia
- Oral intake must be reintroduced SLOWLY to avoid this

NON-PHARMACOLOGICAL MEASURES



Develop a therapeutic alliance and consider family interventions



Nutrition: controlled weight gain, supplementation (e.g. Ensure[®], Boost[®]). Enteral nutrition (e.g. NGT) may be required in severe cases. TPN not recommended; rarely indicated. Consider referral to a dietitian

❖ Hypoglycemia may occur upon resumption of oral intake. May require blood glucose monitoring every 2 hours in the early stages of treatment



Supervised/limited exercise regimen (yoga can be encouraged to reduce anxiety without interfering with rate of weight gain)



Psychotherapy (e.g. CBT, family therapy)

PHARMACOLOGICAL THERAPY

Prokinetic Agents	2 nd Generation Antipsychotics	Benzodiazepines	Nutritional Support	Antidepressants
Domperidone* / Metoclopramide Used to treat gastroparesis by reducing feeling of fullness Prucalopride For constipation and colonic function	Olanzapine Studies show ↓ in delusional thinking and rumination, modest weight gain	E.g. Clonazepam For severe anxiety Caution for dependence	Thiamine/Vitamin B1 given to prevent Wernicke's encephalopathy and refeeding syndrome	Not used routinely SSRIs or SNRIs can be used for co-existing depression, anxiety, purge disorders

*Preferred due to the lower risk of EPS

PROKINETIC AGENTS

DRUG CLASS	SAFETY
Dopamine antagonist -Domperidone	AEs: • Diarrhea • Abdominal discomfort • Drowsiness • Hyperprolactinemia • QT prolongation
Dopamine Antagonist- Metoclopramide	All of above AEs plus: • Restlessness • Rare: EPS
• Indicated for reducing sensation of fullness during early stages of feeding • Domperidone preferred over metoclopramide unless anti-nausea effects needed • Give 30 minutes before meals (domperidone may be taken at bedtime too in some cases) • Domperidone doses >30 mg/day or when used in pts > 60 years of age may be associated with serious cardiac risks	

PROKINETIC AGENTS

DRUG CLASS	SAFETY
5-HT ₄ receptor agonist - Prucalopride	AEs: • Abdominal pain • Nausea • Diarrhea • Headache
• Indicated for treating constipation and normalizing colon function • No known cardiac complications	

2ND GENERATION ANTIPSYCHOTICS

DRUG	SAFETY
Olanzapine	AEs: • Drowsiness • Anticholinergic effects • EPS and akathisia • Dyslipidemia and other metabolic disturbances • DRESS (drug reaction with eosinophilia and systemic symptoms)
• Indicated for psychological symptoms (rumination, delusional thinking) • Treat for ~3 months or until no longer needed for weight gain (max. BMI: 17 kg/m²; continuing after achieving this BMI can cause large increase in appetite and weight gain) • Do NOT use long-term unless benefits outweigh risk of adverse events	

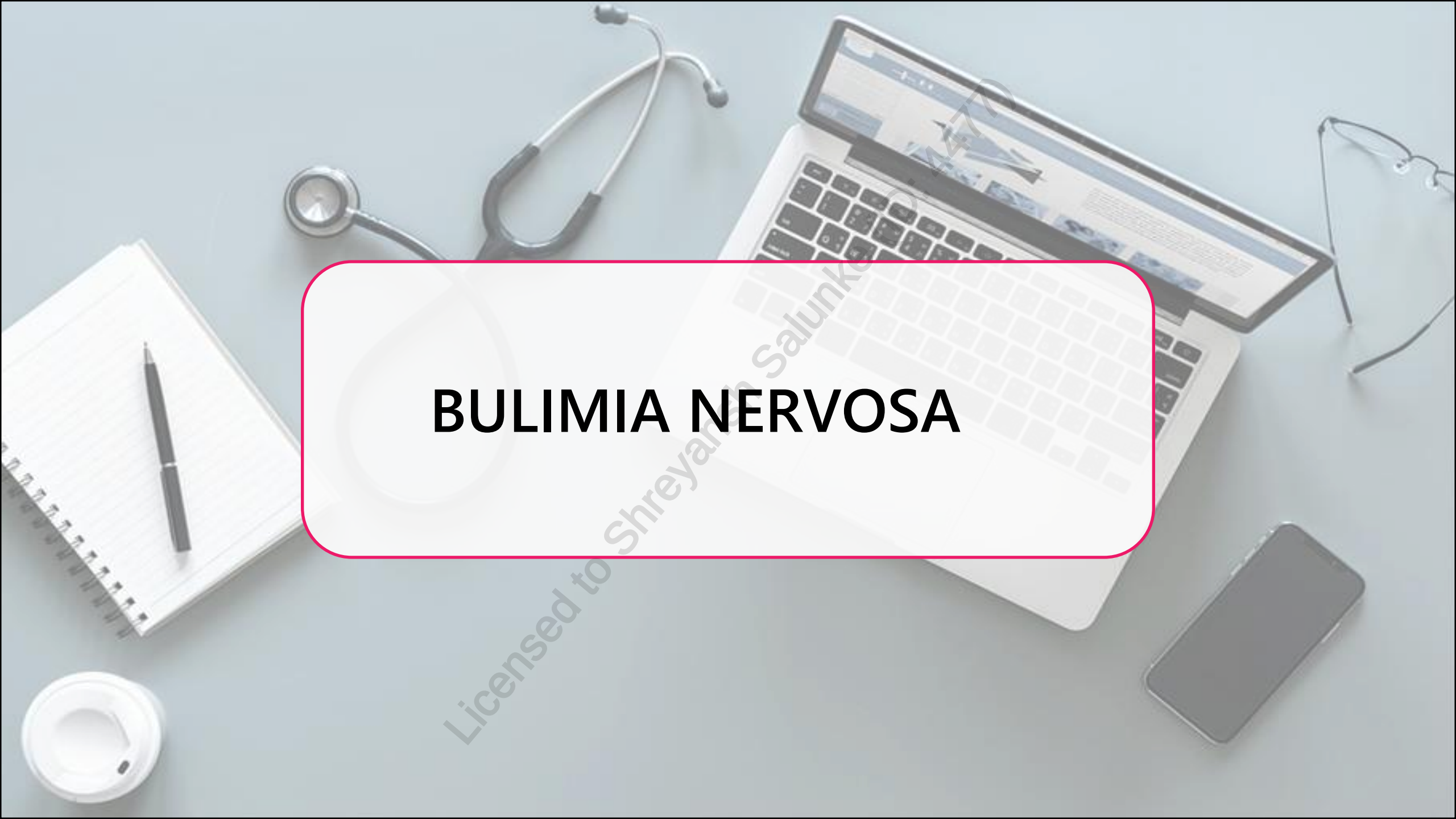
BENZODIAZEPINES

DRUG	SAFETY
Clonazepam, etc.	• Tolerance and/or dependence • Avoid in patients with history of drug/alcohol abuse • Muscle weakness or ataxia • Drowsiness • Memory/cognitive impairment • Risk of falls • Rare: Respiratory depression
• Should be used only for severe concomitant anxiety	

ANOREXIA NERVOSA- MONITORING PARAMETERS

Efficacy Endpoints		
Parameter	Degree of Change	Timeframe
Weight	↑ to normal range	Varies - should generally be done slowly to ↓ risk of refeeding syndrome
Eating habits	Normalize	Ongoing
Psychiatric symptoms	Improve	1-2 months; ongoing therapy may be required
Abnormal lab values (e.g. hypokalemia)	Normalize	1 week-1 month

Safety Endpoints		
Parameter	Degree of Change	Timeframe
AEs of medications	Prevented or minimized	Duration of therapy

A top-down view of a light blue desk with various items: a silver stethoscope, a laptop displaying a medical website, a spiral notebook with a pen, a pair of glasses, a smartphone, and a white coffee cup. A white rounded rectangle with a red border is centered over the laptop.

BULIMIA NERVOSA

BULIMIA NERVOSA- CLINICAL PRESENTATION

Recurrent cycle of binge-eating followed by purging behaviors occurring ≥ 1 x/week for ≥ 3 months

- Binge eating is defined as:
 - Eating significantly more than is considered reasonable given personal and situational factors
 - This occurs within any 2-hour period
 - The patient experiences a sense of loss of control over their eating (they “cannot stop eating”)

Purging behaviors are defined as any of the following:

- Self-induced vomiting, laxative/diuretic misuse, fasting, or excessive exercise

Other symptoms may or may not include:

- Fear & concern over body image
- Noticeable weight fluctuations over time
- Guilt or depression following a binge eating episode
- Low self-esteem
- Menstrual irregularities

BULIMIA NERVOSA- SEVERITY

The DSM-5 classifies severity based on # of episodes of compensatory/purging behaviors per week

- Mild: ~ 1-3 episodes/week
- Moderate: ~ 4-7 episodes/week
- Severe: ~ 8-13 episodes/week
- Extreme: ~ 14+ episodes/week

Other factors must also be considered, including but not limited to:

- Presence of suicidal ideation
- Functional impairment
 - Activities delayed or missed due to bingeing/purging behaviours

BULIMIA NERVOSA- COMPLICATIONS

GENERAL

- Hypotension
- Tachycardia
- Dry skin
- GI problems
- Kidney dysfunction

MISUSE OF LAXATIVES

- Hypokalemia, hyponatremia
- Cathartic colon

SELF-INDUCED VOMITING

- Tooth enamel deterioration
- Tearing of stomach lining
- Esophageal inflammation
- Dehydration
- Hypokalemia, hypochloremia, metabolic alkalosis
- Parotid hypertrophy
- Russell's sign (scars over knuckles)

NON-PHARMACOLOGICAL MEASURES

Non-pharmacological therapy is **1st line therapy** in bulimia nervosa treatment



Psychotherapy (CBT, interpersonal therapy, group therapy)



Monitor for suicidal ideation and treat co-morbid psychiatric conditions



Early intervention through guided self-help approaches (e.g. CBT programs online with clinician coaching) during and post-treatment

PHARMACOLOGICAL THERAPY- ANTIDEPRESSANTS

DRUG CLASS	SAFETY
SSRIs Fluoxetine - 1 st line Other SSRIs - 2 nd line	Headache, GI effects (↑ risk of GI bleeding), anxiety, sexual dysfunction, anticholinergic effects
Venlafaxine	Hypertension, nausea, drowsiness
Trazodone	3 rd line – particularly useful in pts with comorbid insomnia
• Treatment with > 1 antidepressant is not recommended • Purging may ↓ absorption of the medication- timing of administration must be considered • No correlation between co-existing depression and effectiveness of therapy • Do NOT use bupropion (risk of seizures in patients with eating disorders) • Treatment duration: minimum 6 months, ideally at least 1 year • Antidepressants should be tapered down- not stopped abruptly	

MONITORING PARAMETERS

Efficacy Endpoints		
Parameter	Degree of Change	Timeframe
Eating habits	Normalize	Ongoing
Psychiatric symptoms (e.g. anxiety, depression)	Improve	1-2 months
Binge-purge behaviors	Eliminate	1-2 months **
Abnormal lab/vital parameters (e.g. hypokalemia, syncope)	Normalize	1 week-1 month

Safety Endpoints		
Parameter	Degree of Change	Timeframe
AEs of medications	Prevented or minimized	Duration of therapy

**** If long-term misuse of laxatives has occurred, taper slowly over 1 month to 1 year**

A top-down view of a desk with various items: a laptop with a website on the screen, a stethoscope, a spiral notebook with a pen, a pair of glasses, a smartphone, and a white coffee cup. A large white rounded rectangle with a red border is centered over the laptop.

BINGE-EATING DISORDER

Licensed to Shreyans Salunkhe

CLINICAL PRESENTATION

Recurrent episodes of binge-eating occurring ≥ 1 x/week for ≥ 3 months

- Binge eating is defined as:
 - Eating significantly more than is considered reasonable given personal and situational factors
 - This occurs within any 2-hour period
 - The pt experiences a sense of loss of control over their eating (they “cannot stop eating”)

Episodes of binge-eating are NOT associated with recurrent compensatory/purging behaviors

Other symptoms may or may not include:

- Caloric restriction
- Fear and concern over body image
- Noticeable weight fluctuations over time
- Guilt or depression following a binge eating episode
- Low self-esteem
- Menstrual irregularities

SEVERITY

The DSM-5 classifies severity based on # of episodes of binge-eating behaviors per week

- Mild: average 1-3 episodes/week
- Moderate: average 4-7 episodes/week
- Severe: average 8-13 episodes/week
- Extreme: average 14+ episodes/week

Other factors must also be considered, including but not limited to:

- Presence of suicidal ideation
- Functional impairment
 - Activities delayed or missed due to binge-eating behaviours

NON-PHARMACOLOGICAL MEASURES

Non-pharmacological therapy (psychotherapy) is **1st line therapy** in binge-eating disorder



Develop a therapeutic alliance and consider family interventions



Stress management techniques



Psychotherapy (e.g. CBT, family therapy)

- Psychotherapy alone have shown to be more beneficial than pharmacotherapy alone

PHARMACOLOGICAL THERAPY

- Most appropriate as adjunctive to psychotherapy, or as a stand-alone treatment if psychotherapy is inaccessible, ineffective or not preferred

DRUG CLASS	SAFETY
SSRIs Fluoxetine -1 st line Other SSRIs- 2 nd line	Headache, GI effects (↑ risk of GI bleeding), anxiety, sexual dysfunction, anticholinergic effects
Amphetamines (lisdexamfetamine)	Insomnia, irritability, dizziness, headache
• Lisdexamfetamine is the only drug approved by Health Canada for the treatment of binge eating disorders	

MONITORING PARAMETERS

Efficacy Endpoints		
Parameter	Degree of Change	Timeframe
Eating habits	Normalize	Ongoing
Psychiatric symptoms (e.g. anxiety, depression)	Improve	1-2 months
Binge behaviors	Eliminate	1-2 months

Safety Endpoints		
Parameter	Degree of Change	Timeframe
AEs of medications	Prevented or minimized	Duration of therapy

TAKEAWAYS

- There is a lot of overlap between conditions
- Pharmacological therapies should always be used alongside non-pharmacological therapies, not alone
- Resumption of oral intake should occur slowly to avoid refeeding syndrome
- Where possible, family should be involved in treatment, particularly in younger patients
- A trusting relationship with the patient is KEY to effective treatment
- Patients may experience significant psychological distress during bingeing/purging episodes
- Always ask about thoughts of self-harm/suicide
- CBT is the gold standard psychological therapy for eating disorders

CASE STUDY

SY is a 23-year-old female who comes to the clinic with complaints of excessive fine hair growth on her arms and legs, fatigue, frequently feeling cold, dizziness on standing, and the loss of her period after six months of menstrual irregularities. She confirms that she has been very strict with her diet for the past year in an attempt to become healthier, and she sometimes skips meals for fear of gaining any weight. She exercises seven days a week, focusing mainly on aerobic exercises such as running. SY weighs 91 lbs and is 159 cm tall. On further questioning, she mentions that she occasionally gets so hungry that she eats a large amount of food in one sitting and subsequently self-induces vomiting due to feelings of guilt. Her past medical history is insignificant, and she reports no known drug allergies. She does not smoke or drink alcohol.

On physical examination, her blood pressure average is measured as 95/60 mmHg, and her heart rate is 55 beats per minute (bpm).

Laboratory test results returned this morning, revealing a serum potassium level of 3.1 mmol/L, a serum sodium level of 134 mmol/L, a serum calcium level of 2.4 mmol/L, and a serum magnesium level of 0.65 mmol/L. A 12-lead electrocardiogram (ECG) is performed in the physician's office, revealing sinus bradycardia and a corrected QT (QTc) interval of 480 ms (reference <450 ms).



CASE STUDY - QUESTIONS

1. Which of the following is this patient most likely experiencing?
 - a) Anorexia nervosa restricting type
 - b) Anorexia nervosa binge-eating/purging type
 - c) Bulimia nervosa
 - d) Binge-eating disorder
2. How would you classify the severity of SY's anorexia nervosa based on DSM-5 criteria?
 - a) Mild
 - b) Moderate
 - c) Severe
 - d) Extreme
3. Which of the following non-pharmacological options is LEAST appropriate to recommend?
 - a) Moderate-intensity aerobic activity
 - b) Warming therapy
 - c) Cognitive behavioural therapy (CBT)
 - d) Nutritional rehabilitation
4. Which of the following agents is LEAST appropriate to recommend to SY?
 - a) Fluoxetine
 - b) Sertraline
 - c) Fluvoxamine
 - d) Citalopram

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CHANGE LOG

August 2025

- Formatting and grammar changes throughout
- Slide 3: Updated legend
- Slides 5-10, 20-22 & 26-28: Significant changes made throughout
- Slide 10: Updated complications, added dental enamel erosion to complications
- Slide 12: Added 'encephalopathy' after 'Wernicke's'
- Slides 13-16, 23, 29: Removed MOA and Drug Interaction column
- Slide 25: Added Binge Eating Disorders
- Slide 31: Added takeaways slide, removed Clinical Pearls slide and integrated into other slides
- Slides 32-33: Added case study and questions
- Slide 34: Updated references
- Removed slide on erythromycin since it is no longer marketed in Canada

November 2025

- No content changes made

March 2026

- Slide 2: Updated NAPRA competencies
- Slide 3: Updated legend
- Slide 11: Updated non-pharmacological interventions
- Slides 28 and 29: Updated information
- Slide 34: Updated references